

Maintainability

No equipment (system) can be perfectly reliable in spite of the utmost care and best effort on the part of the designer and manufacturer. In fact, very few systems are designed to operate without maintenance of any kind. For a large number of systems, maintenance is a must, as it is one of the effective ways of increasing the reliability of the system.

Usually, two kinds of maintenance are adopted. They are preventive maintenance and corrective or repair maintenance. Preventive maintenance is maintenance done periodically before the failure of the system, so as to increase the reliability of the system by removing the ageing effects of wear, corrosion, fatigue and related phenomena. On the other hand, repair maintenance is performed after the failure has occurred so as to return the system to operation as soon as possible.

The amount and type of maintenance that is used depends on the respective costs and safety consideration of system failure. It is generally assumed that a preventive maintenance action is less costly than a repair maintenance action.

Reliability under preventive Maintenance:-

$$R_M(t) = [R(T)]^n \times R(t - nT)$$

Where

T = Period.

n = no. of maintenance

and $nT < t < (n+1)T$ $n = 0, 1, 2, \dots$

MTTF of a System with preventive maintenance.

$$MTTF = \frac{\int_0^T R(t) dt}{1 - R(T)} = \int_0^\infty R_M(t) dt.$$

Reliability under repair (or) corrective maintenance

Let $M(t)$ denotes the reliability under repair maintenance and $m(t)$ is the pdf of the random variable T .

$$M(t) = \int_0^t m(t) dt$$

$$(MTTR) \text{ Mean time to Repair} = \int_0^{\infty} t m(t) dt$$

Instantaneous repair rate (or) simply the repair

$$\mu(t) = \frac{m(t)}{1 - M(t)}$$

Remarks:-

$$(i) m(t) = M'(t) ;$$

$$(ii) \mu(t) = \frac{M'(t)}{1 - M(t)}$$

$$(iii) M(t) = 1 - e^{-\int_0^t \mu(t) dt}$$

$$(iv) m'(t) = \mu(t) e^{-\int_0^t \mu(t) dt}$$

Pb:1 If a device has a failure rate of

$$\lambda(t) = 0.015 + 0.02t / \text{year}$$

where t is in years.

(a) Calculate the reliability for a 5 year design life, assuming no maintenance is performed.

(b) Calculate the reliability for a 5 year design life, assuming that annual preventive maintenance restores the device to an as good as new condition.

Soln:-

$$(a) R(t) = e^{-\int_0^t \lambda(t) dt} = e^{-\int_0^t (0.015 + 0.02t) dt}$$
$$R(5) = e^{-\int_0^5 (0.015 + 0.02t) dt} = e^{-\left[0.015t + 0.02 \frac{t^2}{2}\right]_0^5}$$
$$= e^{-0.325} = 0.7225$$

(b) We know that $R_n(t) = [R(t)]^n \times R(t-nT)$ after n -maintenance.

In our case $T=1$ $n=4$.

$$\therefore R_n(t) = [R(1)]^4 \times R(t-4)$$

$$R_n(5) = [R(1)]^4 \times R(1) = [R(1)]^5$$
$$= \left[e^{-\int_0^1 [0.015 + 0.02t] dt} \right]^5$$
$$= \left[e^{-0.025} \right]^5 = 0.8825$$

Example:

The time to repair a power generator is best described by its *pdf*

$$m(t) = \frac{t^2}{333}, 1 \leq t \leq 10 \text{ hours}$$

- (a) Find the probability that a repair will be completed in 6 hours.
- (b) What is the MTTR?
- (c) Find the repair rate.

Soln:- (a) $P(T < 6) = P(1 < T < 6) = \int_1^6 m(t) dt$

$$= \int_1^6 \frac{t^2}{333} dt = 0.2152$$

(b) MTTR $= \int_0^{\infty} t m(t) dt = \int_0^{10} \frac{t \cdot t^2}{333} dt = \frac{1}{333} \left[\frac{t^4}{4} \right]_0^{10} = 7.5 \text{ hrs}$

(c) Repair rate $\mu(t) = \frac{m(t)}{1 - M(t)} = \frac{m(t)}{\int_t^{\infty} m(t) dt}$

$$= \frac{t^2/333}{\int_t^{10} \frac{t^2}{333} dt}$$

$$= \frac{t^2/333}{\frac{1}{999} (10^3 - t^3)}$$

$$= \frac{3t^2}{1000 - t^3} \text{ per hour.}$$

$$\left[\begin{aligned} M(t) &= \int_0^t m(t) dt \\ 1 - M(t) &= 1 - \int_0^t m(t) dt \\ &= \int_t^{\infty} m(t) dt \end{aligned} \right]$$